# Polynucleotides

In 1868, Frederich Miescher isolated nucleic acid (then called nuclein) from pus cells. Albrecht Kossel (Nobel prize, 1910) differentiated RNA and DNA in 1882. In 1906, Kossel described the 4 bases in nucleic acids.

Nucleotides are precursors of the nucleic acids, deoxy-ribonucleic acid (DNA) and ribonucleic acid (RNA). The nucleic acids are concerned with the storage and transfer of genetic information. The universal currency of energy, namely ATP, is a nucleotide derivative. Nucleotides are also components of important co-enzymes like $NAD^+$ and FAD, and metabolic regulators such as cAMP and cGMP.

# Composition of Nucleotides

A nucleotide is made up of 3 components:

**a.** *Nitrogenous base*, (a purine or a pyrimidine)

**b.** *Pentose sugar*, either ribose or deoxyribose;

**c.** *Phosphate groups* esterified to the sugar.

When a base combines with a pentose sugar, a **nucleoside** is formed.



Frederich Miescher
1844-1895



Albrecht Kossel
NP 1910
1853-1927



Adolph Strecker
1822-1871



Frederick Hopkins
NP 1929
1861-1947

# Bases Present in the Nucleic Acids

Two types of nitrogenous bases; the purines and pyrimidines are present in nucleic acids.

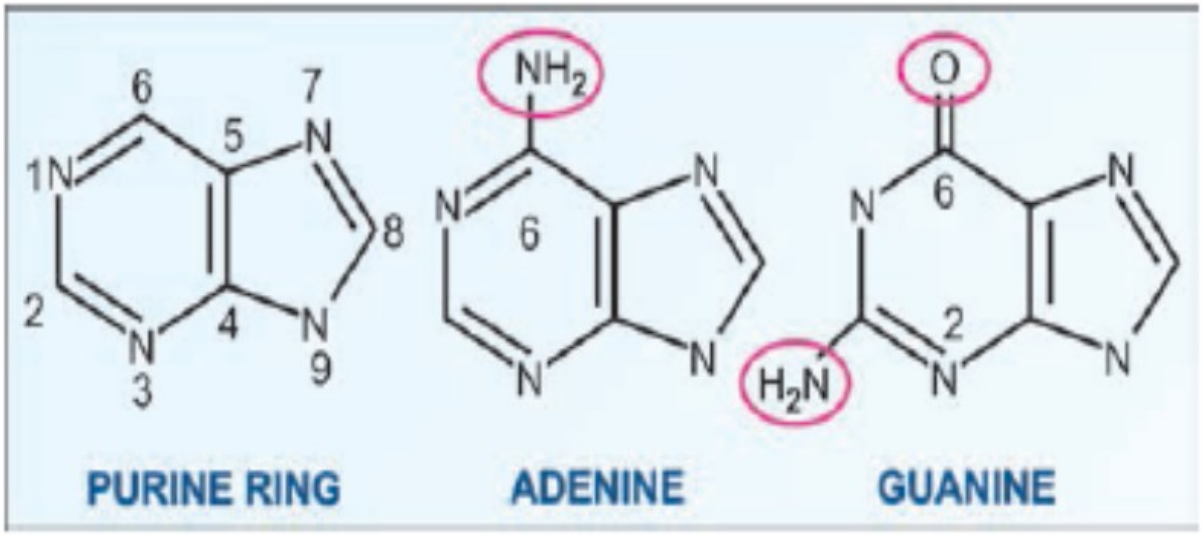



**Fig. 39.1.** Structure of purines

When the nucleoside is esterified to a phosphate group, it is called a **nucleotide** or nucleoside mono-phosphate. When a second phosphate gets esterified to the existing phosphate group, a nucleoside diphosphate is generated. The attachment of a 3rd phosphate group results in the formation of a nucleoside triphosphate. The nucleic acids (DNA and RNA) are polymers of nucleoside monophosphates.

## 1. Purine Bases

The purine bases present in RNA and DNA are the same; **adenine** and **guanine**. Adenine is 6-amino purine and guanine is 2-amino, 6-oxypurine. The numbering of the purine ring with the structure of adenine and guanine are shown in Figure 39.1.

## 2. Minor Purine Bases

These bases may be found in small amounts in nucleic acids and hence called minor bases. These are **hypoxanthine** (6-oxopurine) and **xanthine** (2, 6-di-oxopurine) (Fig. 39.2). **Uric acid** (2,6,8-tri-oxopurine) is formed as the end product of the catabolism of other purine bases. It can exist in the "enol" as well as "keto" forms (tautomeric forms) (Fig. 39.2). Keto form is by far the predominant type under physiological conditions.

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## 3. Pyrimidine Bases

The pyrimidine bases present in nucleic acids are cytosine, thymine and uracil. **Cytosine** is present in both DNA and RNA. **Thymine** *is present in DNA and* **uracil** *in RNA*. Structures are shown in Figure 39.3. See also Box 39.1.

A few other modified pyrimidine bases like dihydrouracil and 5-methyl cytosine are also found rarely in some types of RNA (Fig. 39.4).

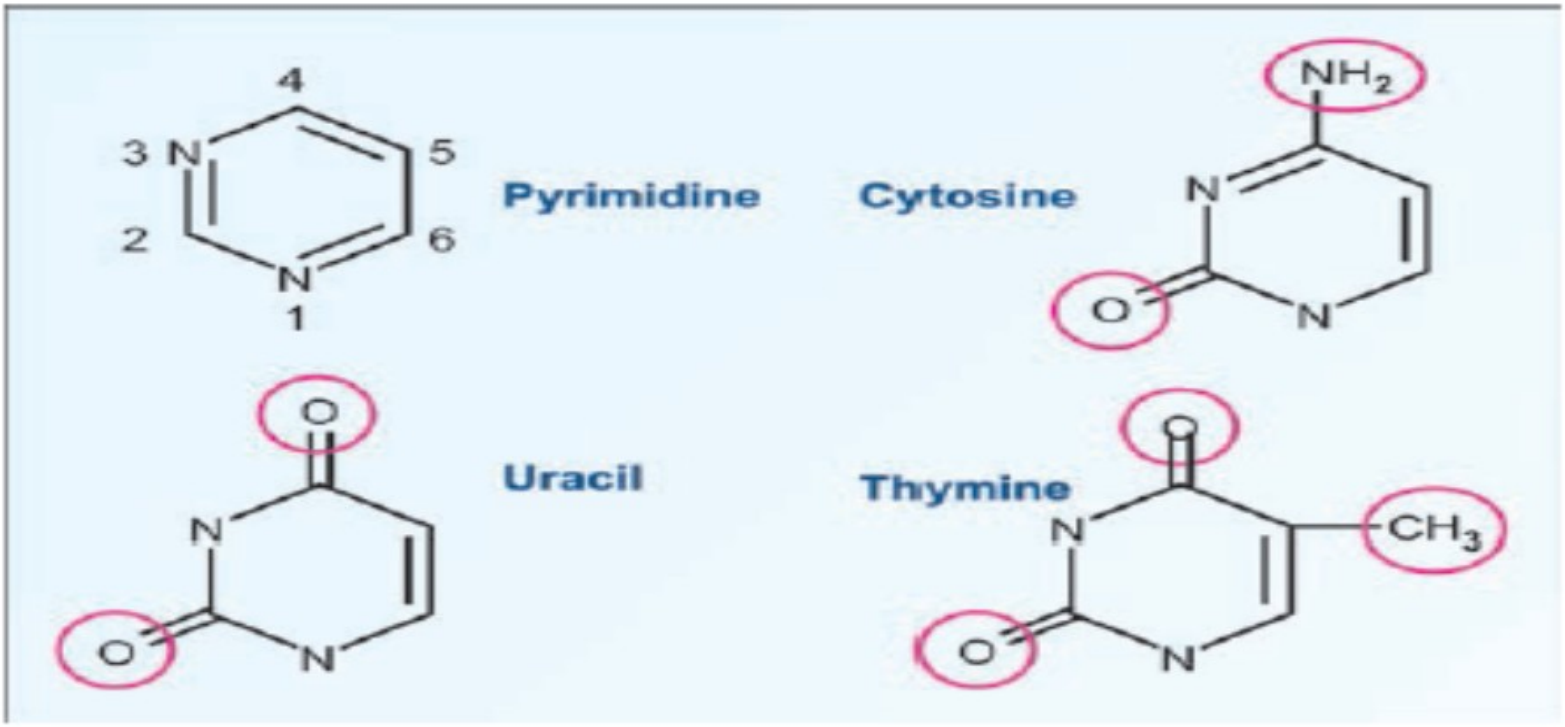



**Fig. 39.3.** Common pyrimidines

## Box 39.1. These Two Words are Often Confused

**THYMINE** is the base present in DNA

**THIAMINE** is a member of vitamin B complex

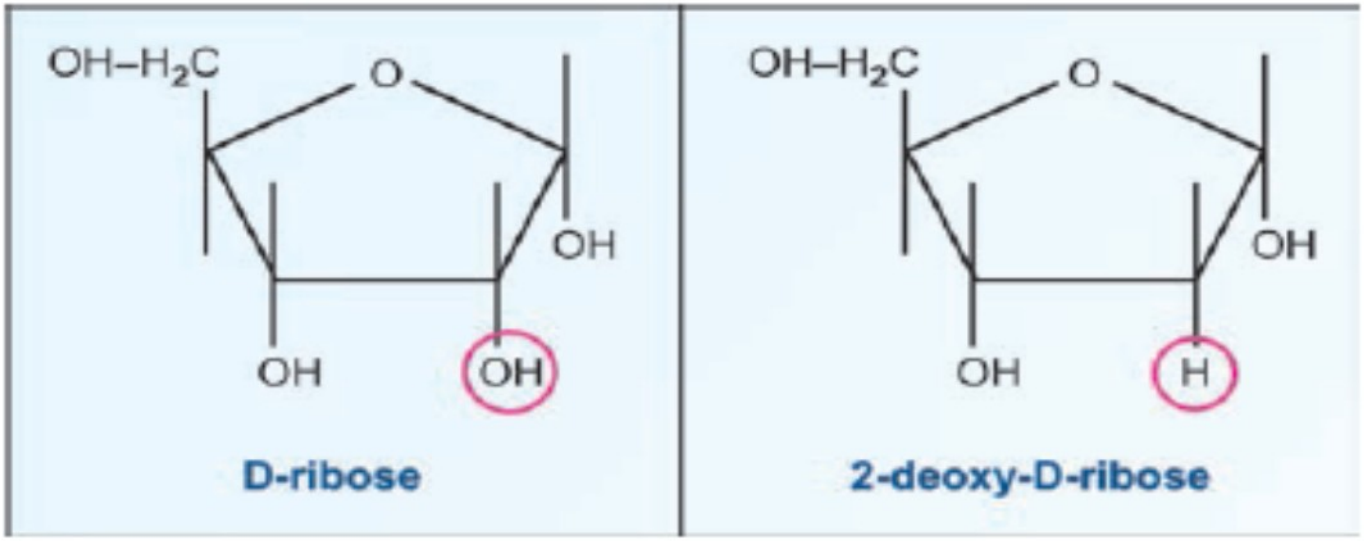



**Fig. 39.5.** Sugar groups in nucleic acids

## Nucleosides

i. Nucleosides are formed when bases are attached to the pentose sugar, D-ribose or 2-deoxy D-ribose (Fig. 39.5).

ii. All the bases are attached to the corresponding pentose sugar by a beta-N-glycosidic bond between the 1st carbon of the pentose sugar and N9 of a purine or N1 of a pyrimidine.

iii. *The deoxy nucleosides are denoted by adding the prefix d- before the nucleoside.*

iv. The carbon atoms of the pentose sugar are denoted by using a prime number to avoid confusion with the carbon atoms of the purine or pyrimidine ring (Fig. 39.6). The names of the different nucleosides are given in Table 39.1.

v. Nucleosides with purine bases have the suffix **-sine**, while pyrimidine nucleosides end with **-dine**.

vi. *Uracil combines with ribose only; and thymine with deoxy ribose only* (Table 39.1).

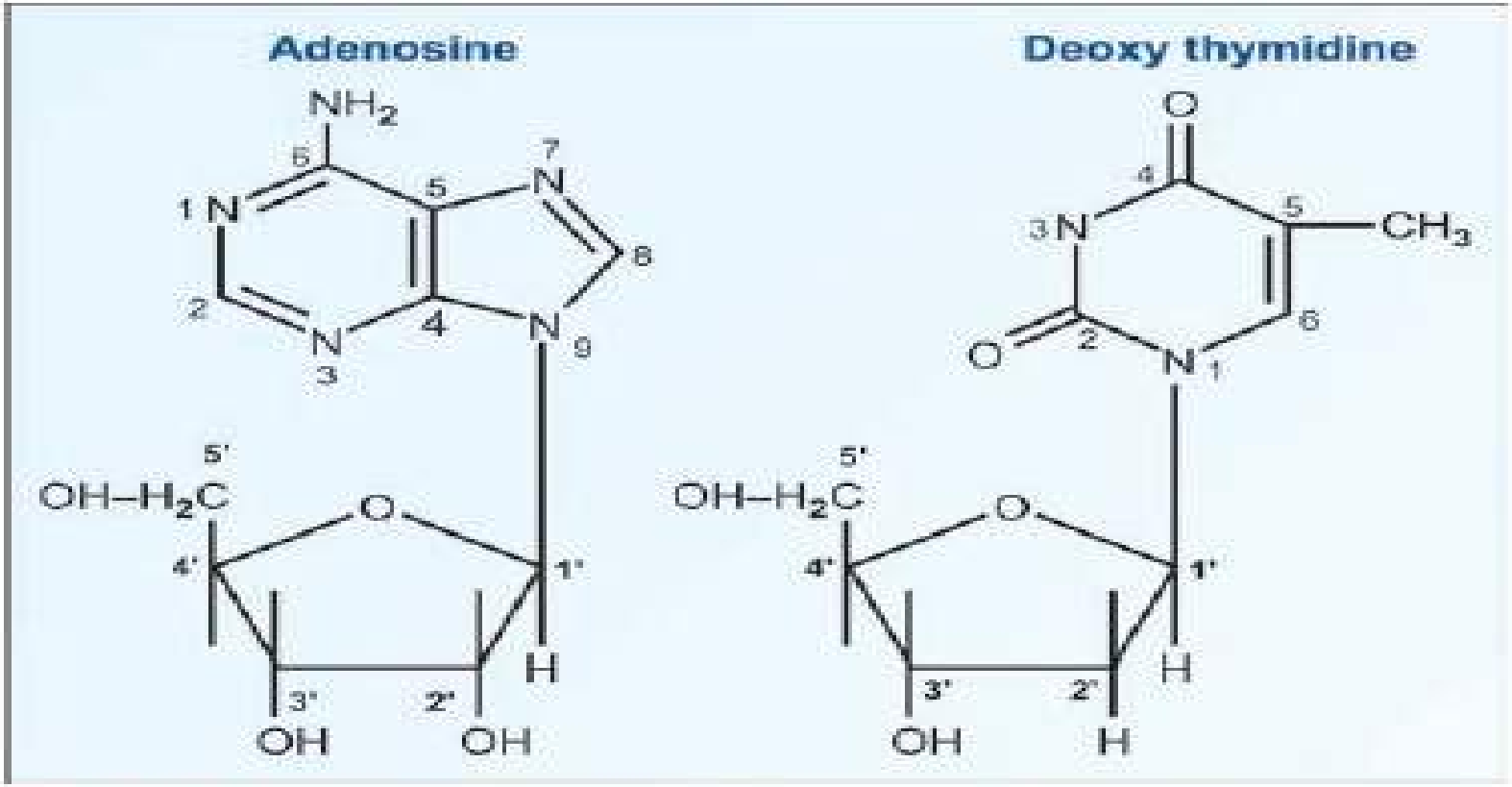



**Fig. 39.6.** Numbering in base and sugar groups. Atoms in sugar is denoted with primed numbers

## Table 39.1. Base + sugar are nucleosides

| Ribonucleosides | | | | |
|---|---|---|---|---|
| **Adenine** | **+** | **Ribose** | **→** | **Adenosine** |
| **Guanine** | **+** | **Ribose** | **→** | **Guanosine** |
| **Uracil** | **+** | **Ribose** | **→** | **Uridine** |
| **Cytosine** | **+** | **Ribose** | **→** | **Cytidine** |
| **Hypoxanthine** | **+** | **Ribose** | **→** | **Inosine** |
| **Xanthine** | **+** | **Ribose** | **→** | **Xanthosine** |
| Deoxy ribonucleosides | | | | |
| Adenine | + | Deoxy ribose | → | Deoxy adenosine (d-adenosine) |
| Guanine | + | Deoxy ribose | → | d-guanosine |
| Cytosine | + | Deoxy ribose | → | d-cytidine |
| Thymine | + | Deoxy ribose | → | d-thymidine |

## Nucleotides

i. These are phosphate esters of nucleosides. Base plus pentose sugar plus phosphoric acid is a nucleotide.

ii. The esterification occurs at the 5th or 3rd hydroxyl group of the pentose sugar. Most of the nucleoside phosphates involved in biological function are 5'-phosphates (Table 39.2).

iii. Since 5'-nucleotides are more often seen, they are simply written without any prefix. For example, 5'-AMP is abbreviated as AMP; but 3' variety is always written as 3'-AMP.

iv. Moreover, a base can combine with either ribose or deoxy ribose, which in turn can be phosphorylated at 3' or 5' positions. One purine and one pyrimidine derivative are given as examples in Table 39.3.

v. Many co-enzymes are derivatives of adenosine monophosphate. Examples are $NAD^+$, NADP, FAD and Co-enzyme A.

vi. Nucleotides and nucleic acids absorb light at a wavelength of 260 nm; this aspect is used to quantitate them. As nucleic acids absorb ultraviolet light, chemical modifications are produced leading to mutation and carcinogenesis.

## Table 39.2. Base+sugar+phosphate = nucleotide

| | | |
|---|---|---|
| **Ribonucleotides** | | |
| Adenosine | + Pi | →Adenosine monophosphate (AMP) (Adenylic acid) |
| Guanosine | + Pi | →Guanosine monophosphate (GMP) (Guanylic acid) |
| Cytidine | + Pi | →Cytidine monophosphate (CMP) (Cytidylic acid) |
| Uridine | + Pi | →Uridine monophosphate (UMP) (Uridylic acid) |
| Inosine | + Pi | →Inosine monophosphate (IMP) (Inosinic acid) |
| **Deoxy ribonucleotides** | | |
| d-adenosine | + Pi | →d-AMP (d-adenylic acid) |
| d-guanosine | + Pi | →d-GMP (d-guanylic acid) |
| d-cytidine | + Pi | →d-CMP (d-cytidylic acid) |
| d-thymidine | + Pi | →d-TMP (d-thymidylic acid) |

## Table 39.3. Nucleosides and nucleotides

| Base | Sugar | Nucleoside | Phospho-ric acid at | Nucleotide |
|---|---|---|---|---|
| **Adenine** | ribose | adenosine | 5' position | AMP |
| do | do | do | 3' position | 3'-AMP |
| do | deoxy-ribose | d-adeno-sine | 5' position | d-AMP |
| do | do | do | 3' position | d-3'-AMP |
| **Cytosine** | ribose | cytidine | 5' position | CMP |
| do | do | do | 3' position | 3'-CMP |
| do | deoxy-ribose | d-cytidine | 5' position | d-CMP |
| do | do | do | 3' position | d-3'-CMP |

## Nucleoside triphosphates

i. Corresponding nucleoside di- and tri- phosphates are formed by esterification of further phosphate groups to the existing ones. In general, any nucleoside triphosphate is abbreviated as NTP or d-NTP (See Table 39.4).

ii. Nucleoside diphosphate contains one high energy bond and triphosphates have 2 high energy bonds. ATP is the **universal energy currency** (Fig. 39.7). It is formed during oxidative processes by trapping the released energy in the high energy phosphate bond. More details on high energy bonds are given in Chapter 19.

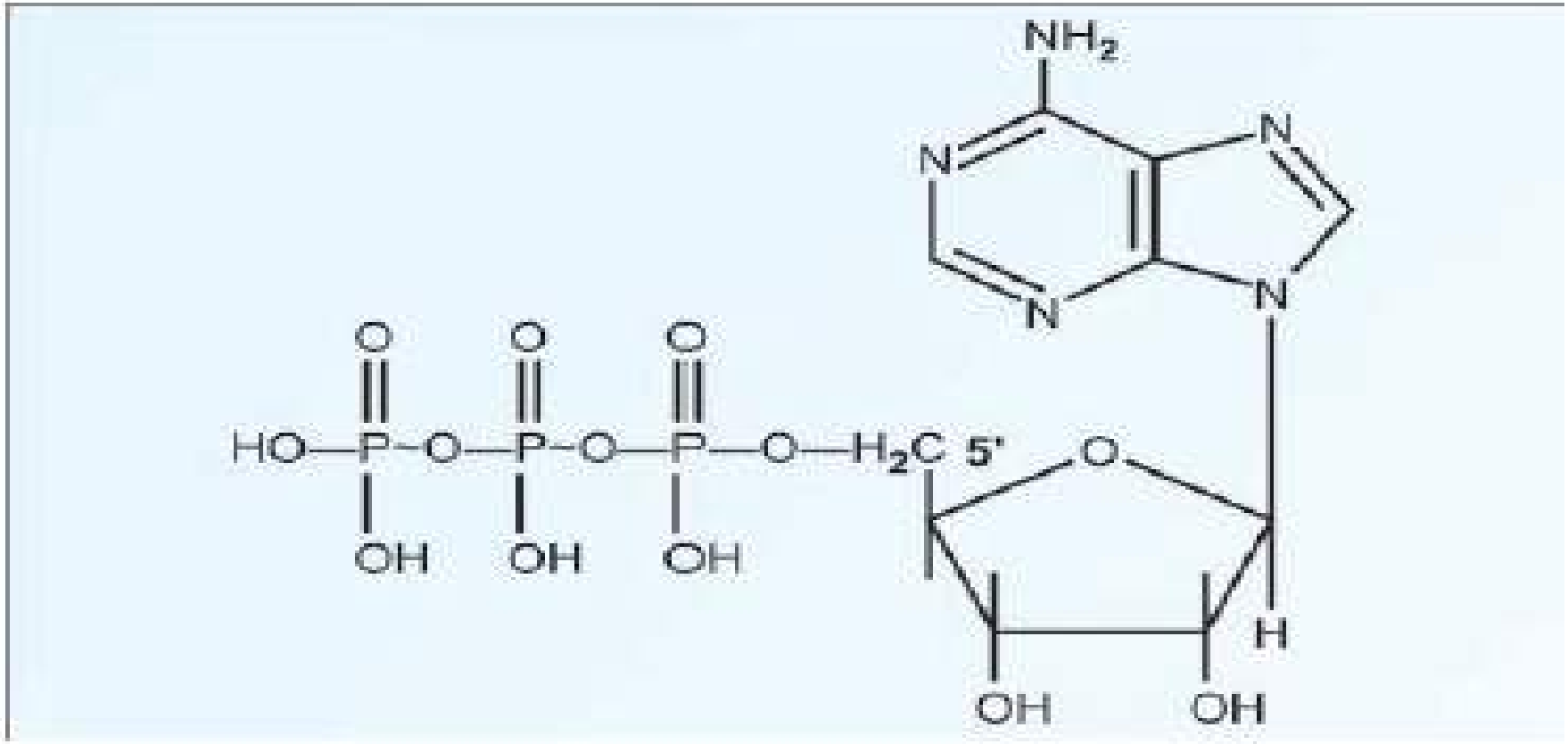



**Fig. 39.7.** Adenosine triphosphate (ATP)

## Table 39.4. Nucleoside triphosphates

| Nucleoside | Nucleoside monophos-phate | Nucleoside diphos- phate(NDP) | Nucleoside triphos-phate (NTP) |
|---|---|---|---|
| **Ribonucleoside phosphates** | | | |
| **Adenosine** | Adenosine mono-phosphate **(AMP)** | Adenosine diphosphate **(ADP)** | Adenosine triphosphate **(ATP)** |
| **Guanosine** | **GMP** | **GDP** | **GTP** |
| **Inosine** | **IMP** | **IDP** | **ITP** |
| **Cytidine** | **CMP** | **CDP** | **CTP** |
| **Uridine** | **UMP** | **UDP** | **UTP** |
| **Deoxy ribonucleoside phosphates** | | | |
| **d-adenosine** | **d-AMP** | **d-ADP** | **d-ATP** |
| **d-guanosine** | **d-GMP** | **d-GDP** | **d-GTP** |
| **d-cytidine** | **d-CMP** | **d-CDP** | **d-CTP** |
| **d-thymidine** | **d-TMP** | **d-TDP** | **d-T T P** |

iii. A phosphodiester linkage may be formed between the 3' and 5' positions of ribose group. Such compounds are called cyclic nucleotides (Fig. 39.8). 3', 5'-**cyclic AMP** or cAMP is a major metabolic regulator. Cyclic GMP also behaves similarly. These are second messengers in mediating the action of several hormones.

iv. Deoxy ribonucleotides are used for synthesis of DNA and ribonucleotides for RNA. In pseudo-uridylic acid (found in tRNA) uridine is attached to ribose phosphate in a C-C bond instead of C-N bond in UMP (Fig. 39.9).

v. High energy compounds are listed in Table 19.2. Please note that active methionine, amino acid adenylates, active sulfate etc are higher energy compounds containing adenosine monophos-phate.

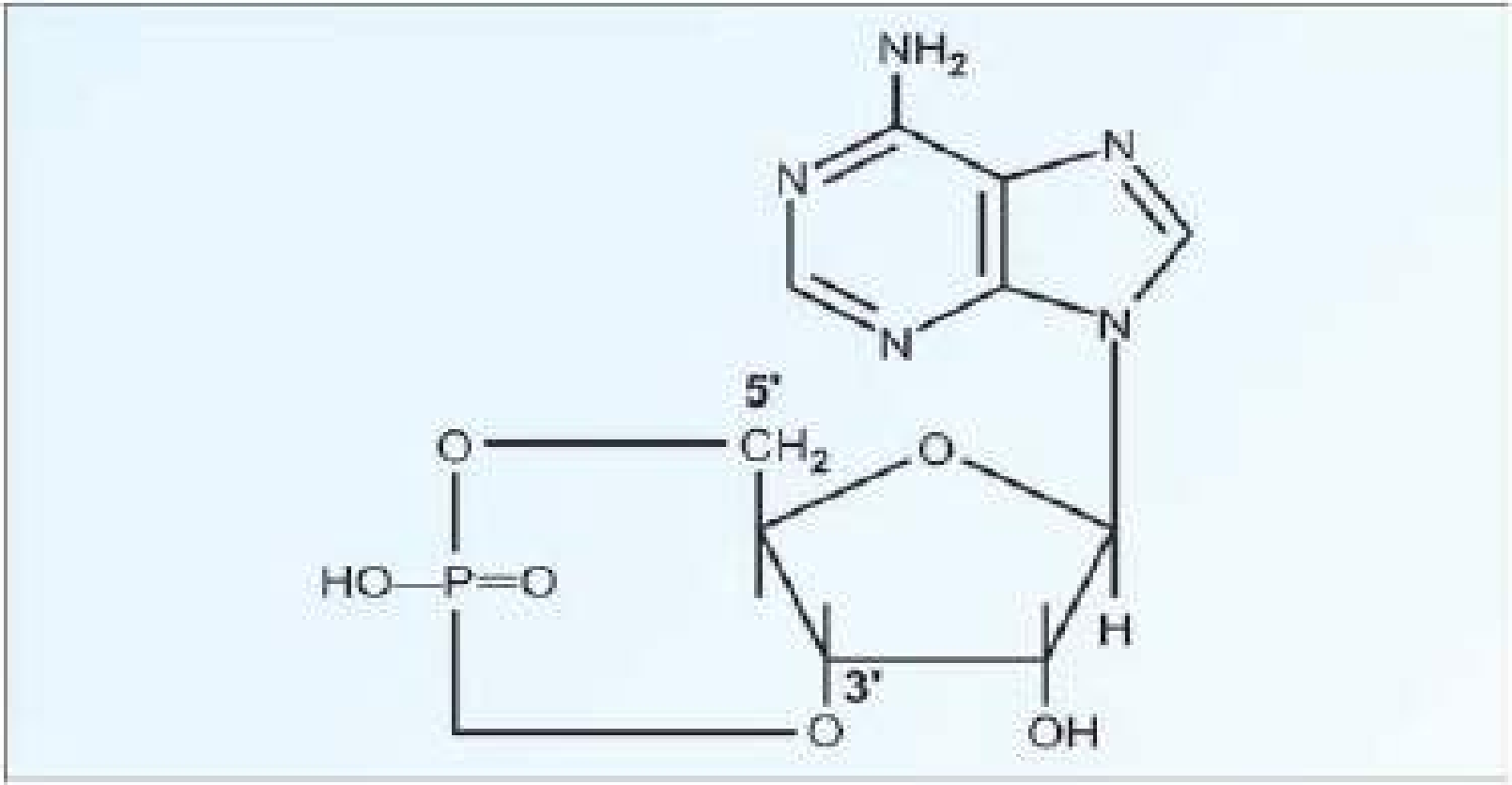



**Fig. 39.8.** 3',5'-cyclic AMP or cAMP

## Digestion of Nucleic Acids

The nucleic acids in the diet are hydrolyzed to a mixture of nucleotides by ribonuclease and deoxy ribonuclease present in pancreatic and intestinal secretions. Then nucleotidases liberate the phosphate from nucleotides. The resulting nucleosides are hydrolyzed by nucleosidases forming free bases and pentose sugars.

However, the dietary purines and pyrimidines are neither converted to nucleotides nor incorporated into nucleic acids. They are directly catabolized.

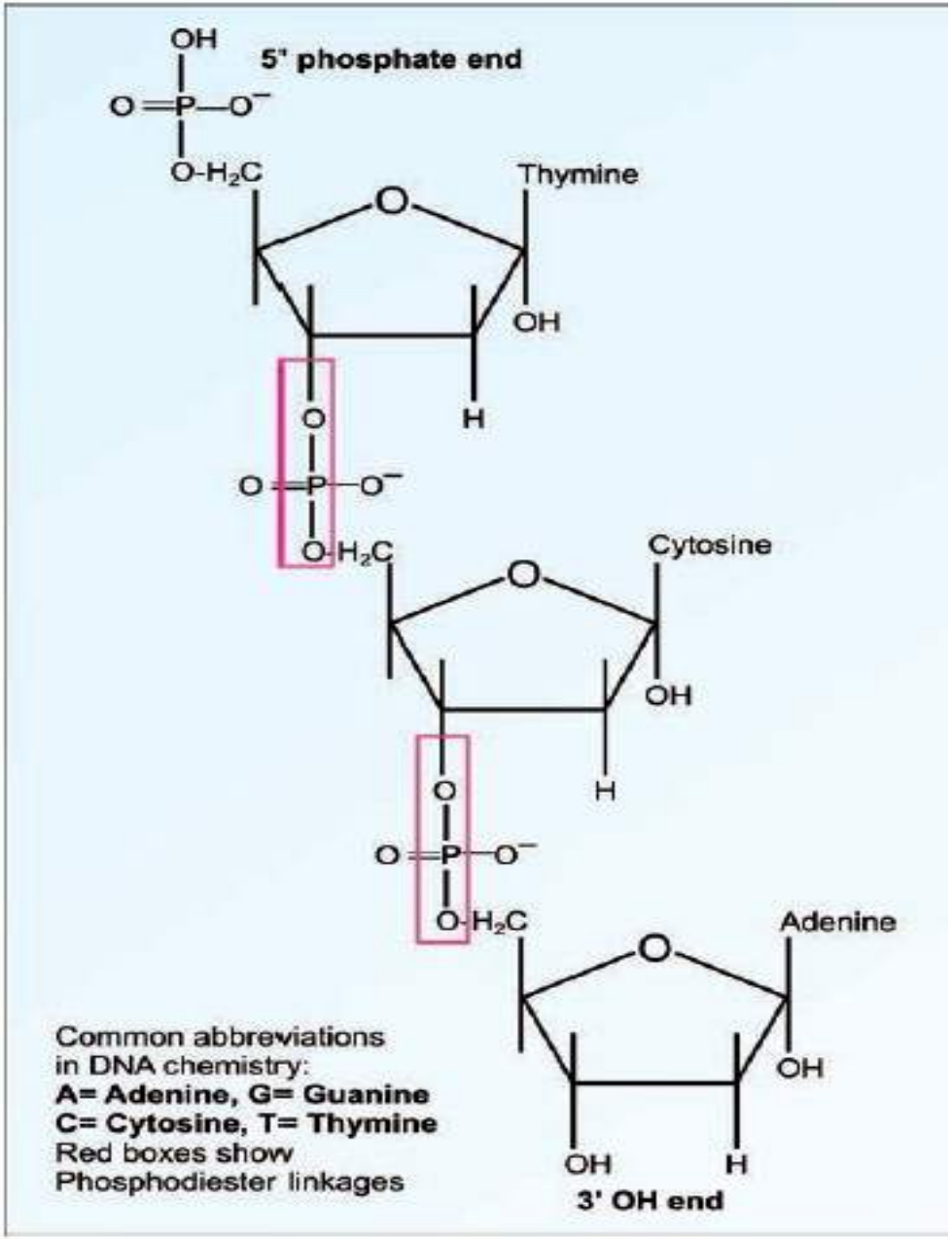


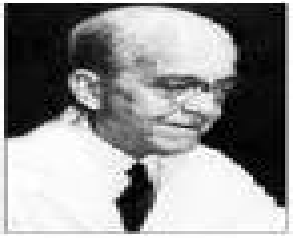

Fig. 40.1. Polynucleotide

Thomas Morgan (1866-1945), the founder of modern genetics, showed that chromosomes contain genes in a sequential manner in Drosophila (Nobel prize, 1933). In 1931, Barbara McClintock showed the rearrangement of genes or mobile genes in chromosomes in corn (Nobel prize, 1983). George Beadle, working with mutant strains of Neurospora suggested "one enzyme one gene" hypothesis in 1941 (Nobel prize, 1958). Avery in 1944 demonstrated that DNA is the genetic material. In 1952, Hershey (Nobel Prize 1946) showed that only DNA of virus and not the proteins will enter into the host before infection. Edwin Chargaff elicited the base pairing rule of DNA in 1950. X-ray crystallographic studies on DNA by Maurice Wilkins (Nobel prize, 1962) showed the details of structure of DNA. Rosalind Franklin worked out the helical structure of DNA. (She died of ovarian cancer at the age of 37, probably due to irradiation during her work on X-ray crystallography). Based on these data, James Watson and Francis Crick in 1953 deduced the double helical structure of DNA (Nobel prize, 1962).

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Thomas Morgan
NP 1933
1866-1945



Oswald Avery
1877-1955



Alfred Hershey
NP 1969
1908-1997



George Beadle
NP 1958
1903-1989



Barbara McClintock
NP 1983
1902-1992

## STRUCTURE OF DNA

Deoxyribonucleic acid (DNA) is composed of four deoxy ribonucleotides, i.e. deoxyadenylate (A), deoxyguanylate (G), deoxycytidylate (C), and thymidylate (T).

These units are combined through 3' to 5' **phospho diester bonds** to polymerise into a long chain. The nucleotide is formed by a combination of base + sugar + phosphoric acid. The 3'-hydroxyl of one sugar is combined to the 5'-hydroxyl of another sugar through a phosphate group (Fig. 40.1). In this particular example, the thymidine is attached to cytidine and then cytidine to adenosine through phospho-diester linkages (Fig. 40.1).

In the DNA, the base sequence is of paramount importance. The genetic information is coded in the

specific **sequence of bases**; if the base is altered, the information is also altered.

The deoxyribose and phosphodiester linkages are the same in all the repeating nucleotides. Therefore, the message will be conveyed, even if the base sequences alone are mentioned as shown:

5'P--Thymine--Cytosine-Adenine-3'OH
Or, 5'-----T--C--A---3'

This would convey all the salient features of the polynucleotide shown in Figure 40.1.

## Polarity of DNA molecule

In the case of DNA, the base sequence is always written from the 5' end to the 3' end. This is called the polarity of the DNA chain.

## Watson-Crick Model of DNA Structure

The salient features of Watson-Crick model of DNA are given below (Figs 40.2 and 40.3):

### 1. Right handed double helix

DNA consists of two polydeoxy ribonucleotide chains twisted around one another in a right handed double helix similar to a spiral stair case. The sugar and phosphate groups comprise the handrail and the bases jutting inside represent the steps of the

staircase. The bases are located perpendicular to the helix axis, whereas the sugars are nearly at right angles to the axis.

## 2. The base pairing rule

Always the two strands are **complementary** to each other. So, the adenine of one strand will pair with thymine of the opposite strand, while guanine will pair with cytosine. The base pairing (**A with T; G with C**) is called **Chargaff's rule**, which states that the number of purines is equal to the number of pyrimidines.

## 3. Hydrogen bonding

The DNA strands are held together mainly by hydrogen bonds between the purine and pyrimidine bases. There are two hydrogen bonds between A and T while there are three hydrogen bonds between C and G. The GC bond is therefore stronger than the AT bond.

## 4. Antiparallel

The two strands in a DNA molecule run antiparallel, which means that one strand runs in the 5' to 3' direction, while the other is in the 3' to 5' direction. (Fig. 40.2). This is similar to a road divided into two, each half carrying traffic in the opposite direction.

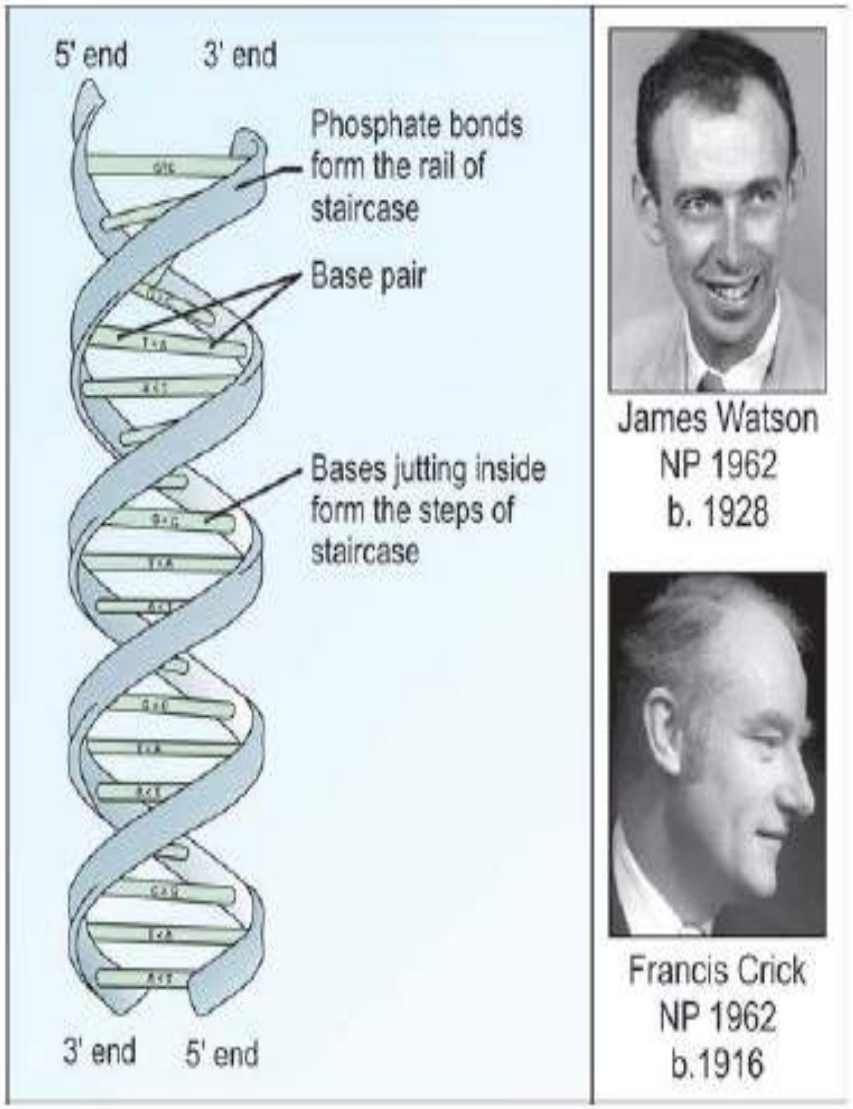



**Fig. 40.2.** Watson-Crick model of double helical structure of DNA. Adjacent bases are separated by 0.34 nm. The diameter or width of the helix is 2 nanometers

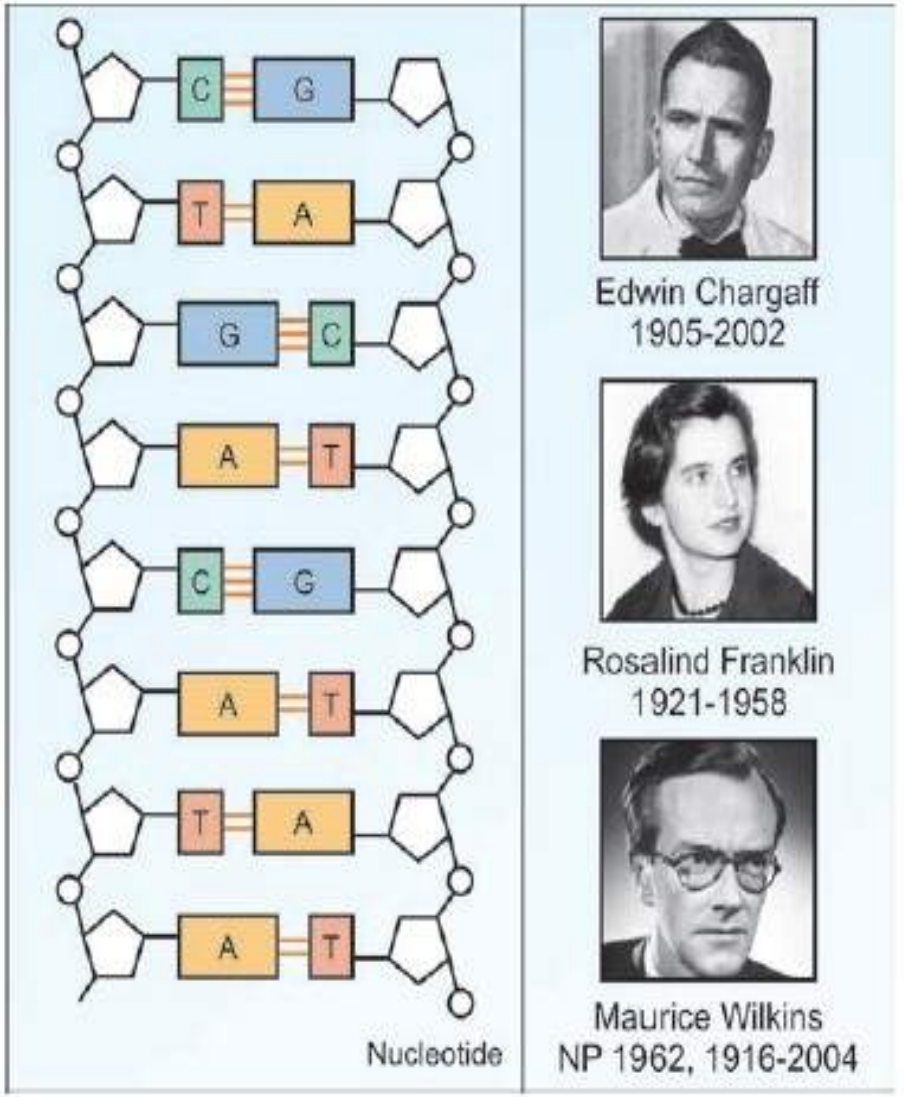



**Fig. 40.3.** Base pairing rule. Base pairing of A with T and G with C. Hydrogen bonds between bases

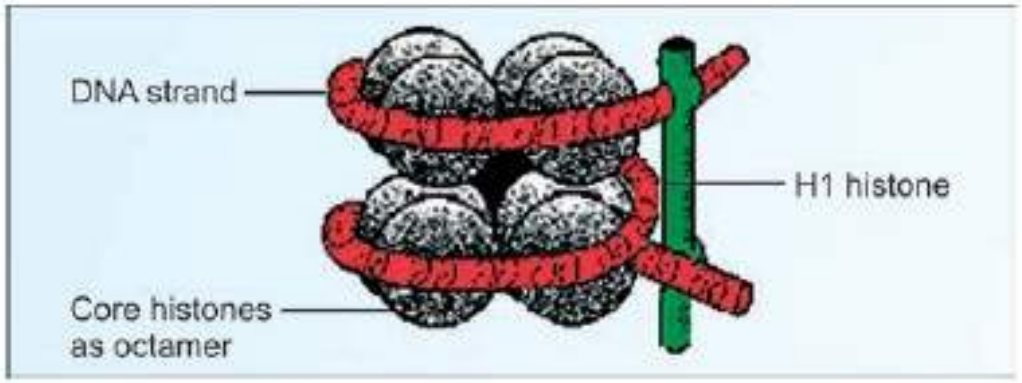



**Fig. 40.4.** DNA wraps twice around histone octamer to form one nucleosome

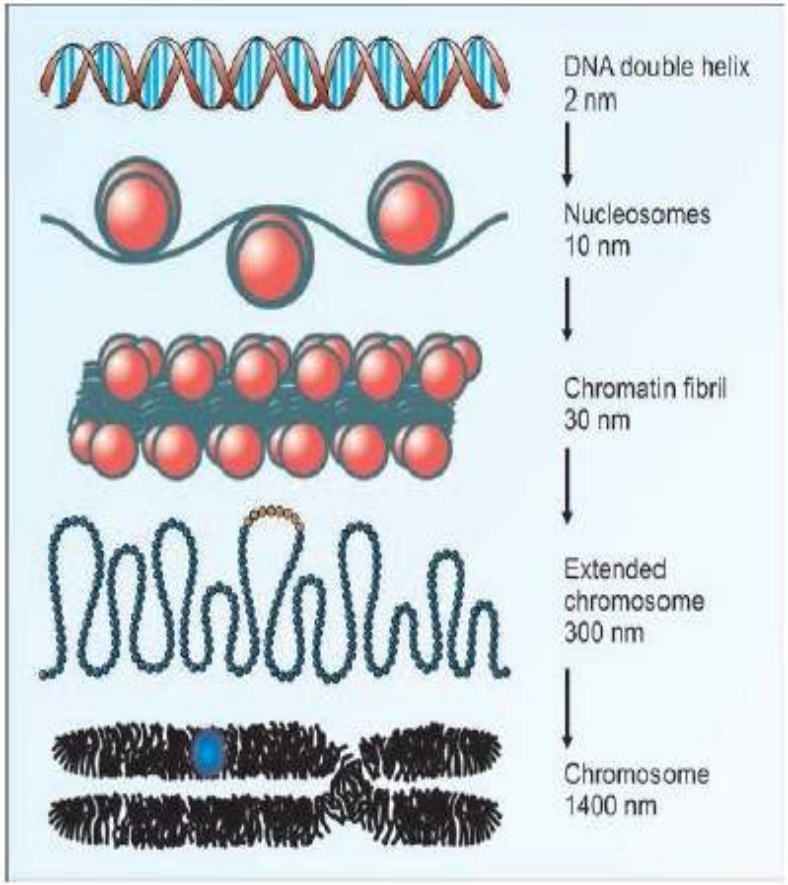



**Fig. 40.5.** DNA condenses repeatedly to form chromosome

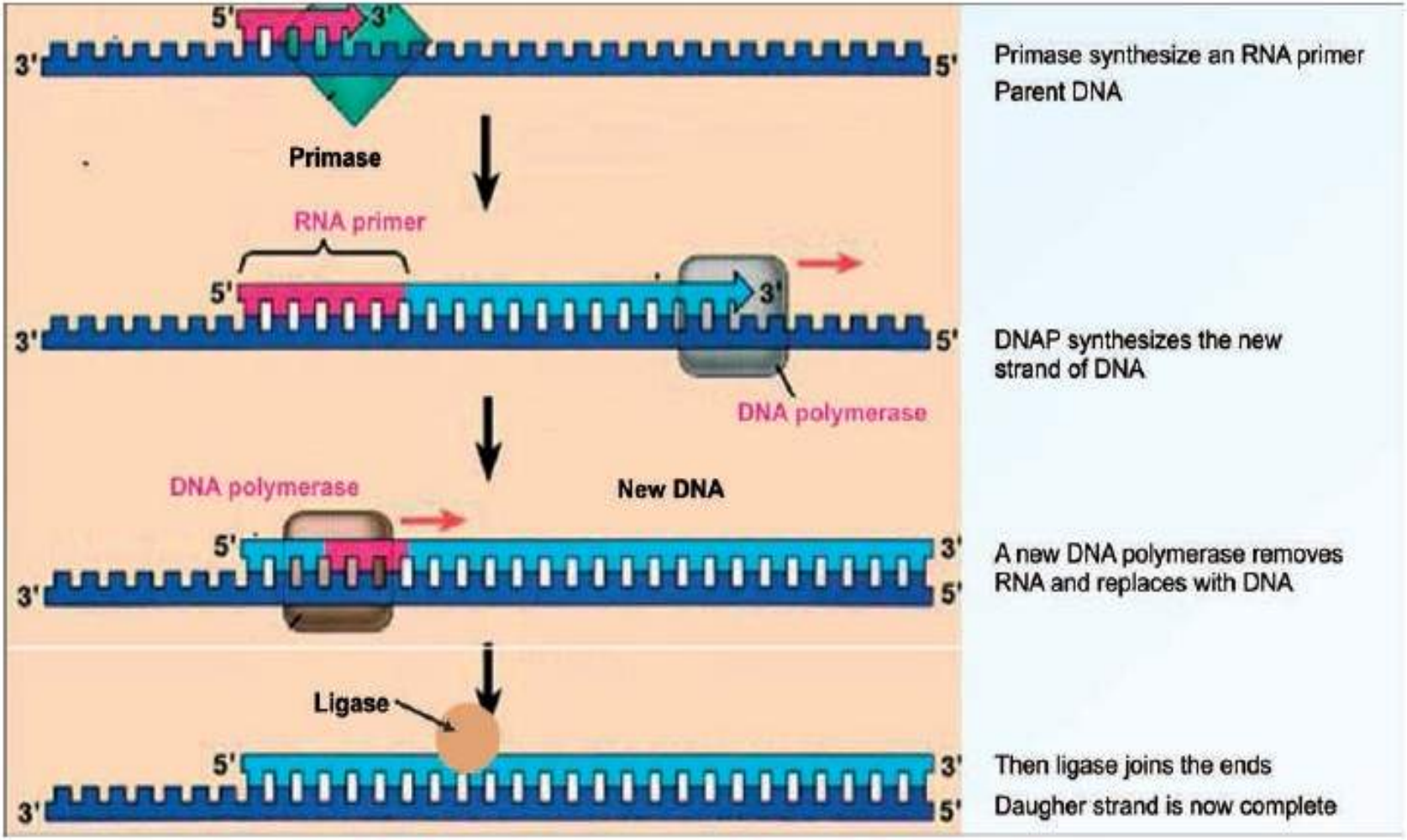
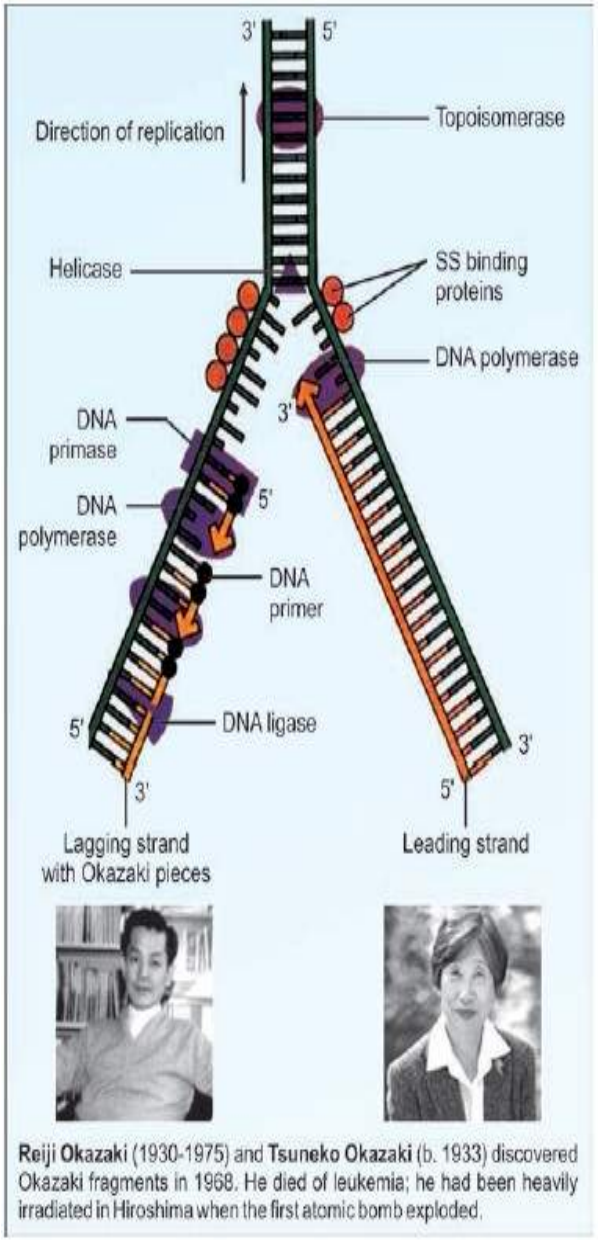



**Fig. 40.13.** RNA primer is needed for the DNA synthesis




**Reiji Okazaki** (1930-1975) and **Tsuneko Okazaki** (b. 1933) discovered Okazaki fragments in 1968. He died of leukemia; he had been heavily irradiated in Hiroshima when the first atomic bomb exploded.

**Fig. 40.14.** Lagging strand and Okazaki pieces

## **Box 40.1.** Summary of DNA Replication

1. Origin of replication is identified. Then unwinding of parental DNA to form a replication fork.
2. RNA primer complementary to the DNA template is synthesized by RNA primase.
3. DNA synthesis is continuous in the leading strand (towards replication fork) by DNA polymerase.
4. DNA synthesis is discontinuous in the lagging strand (away from the fork), as Okazaki fragments.
5. Elongation: In both strands, the synthesis is from 5' to 3' direction.
6. Then the RNA pieces are removed; the gaps filled by deoxynucleotides by DNAP and the pieces are ligated by DNA ligase.
7. Proof reading is done by the DNA polymerase.
8. Finally organised into chromatin.
9. Main enzymes involved in replication are: DNA polymerases; Helicases; Topoisomerases; DNA primase; Single strand binding proteins; and DNA ligase.

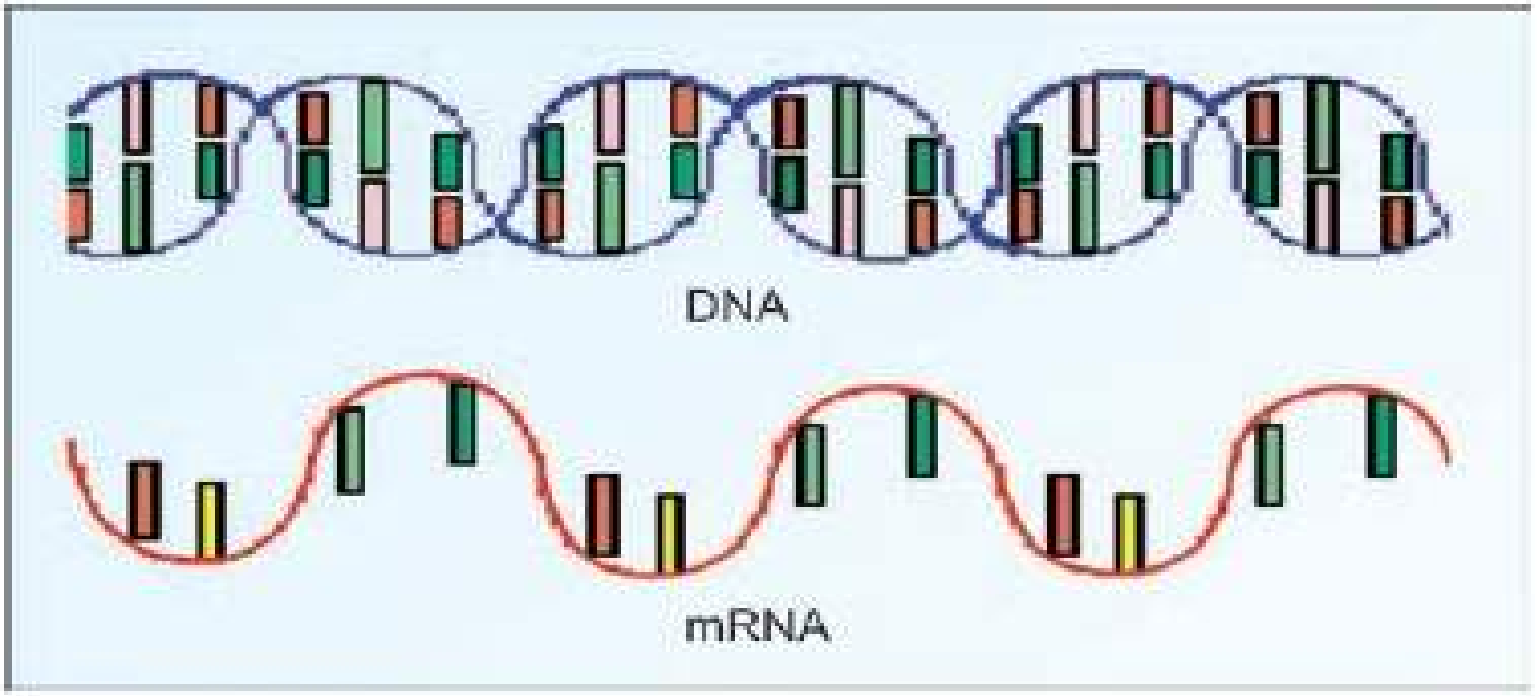



**Fig. 41.1.** DNA is double stranded; while RNA single stranded

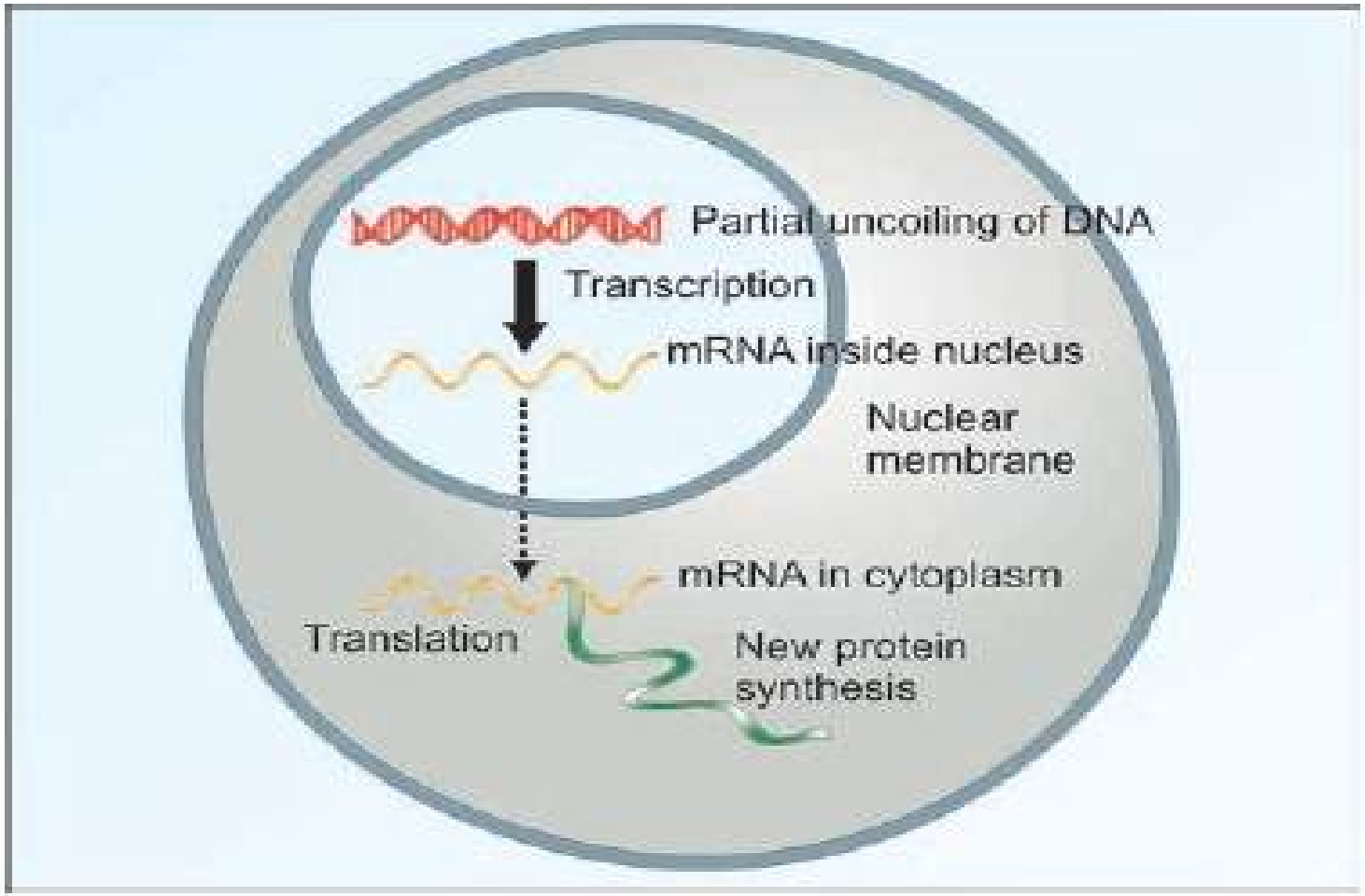



**Fig. 41.2.** Central dogma of molecular biology

# Types of RNA

a. Messenger RNA (mRNA). The gene present in DNA is transcribed into mRNA. This constitutes about 2-5% of total RNA in the cell. They are generally degraded quickly.
b. Ribosomal RNA (rRNA). This constitutes about 80% of all RNA in the cell. 28S, 18S and 5S are the major varieties. They are involved in the protein biosynthesis. They are very stable.
c. Transfer RNA (tRNA). There are about 60 different species present. They constitute about 15% of the total RNA in the cell. They are very stable.
d. Small RNA. They constitute about 1-2% of total RNA in the cell. There are about 30 different varieties. They are very stable. **Small Nuclear RNAs** (SnRNAs) are a subgroup of small RNA. Some important species of SnRNAs are U1 (165 nucleotides), U2 (188 nucleotides), U3 (216), U4 (139), U5 (118), U6 (106). They are involved in mRNA splicing.
e. Micro-RNA (miRNA). They alter the function of mRNA. They are moderately stable. More details are given at the end of this chapter.

**Table 41.1.** Differences between RNA and DNA

| RNA | DNA |
|---|---|
| 1. Mainly seen in **cytoplasm** | Mostly inside **nucleus** |
| 2. Usually 100-5000 bases | Millions of base pairs |
| 3. Generally single stranded | Double stranded |
| 4. Sugar is **ribose** | Sugar is **deoxyribose** |
| 5. Purines: Adenine, Guanine Pyrimidines: Cytosine, **Uracil** | Adenine, Guanine Cytosine, **Thymine** |
| 6. Guanine content is not equal to cytosine and adenine is not equal to uracil | Guanine is equal to cytosine and adenine is equal to thymine |
| 7. Easily destroyed by alkali | **Alkali resistant** |

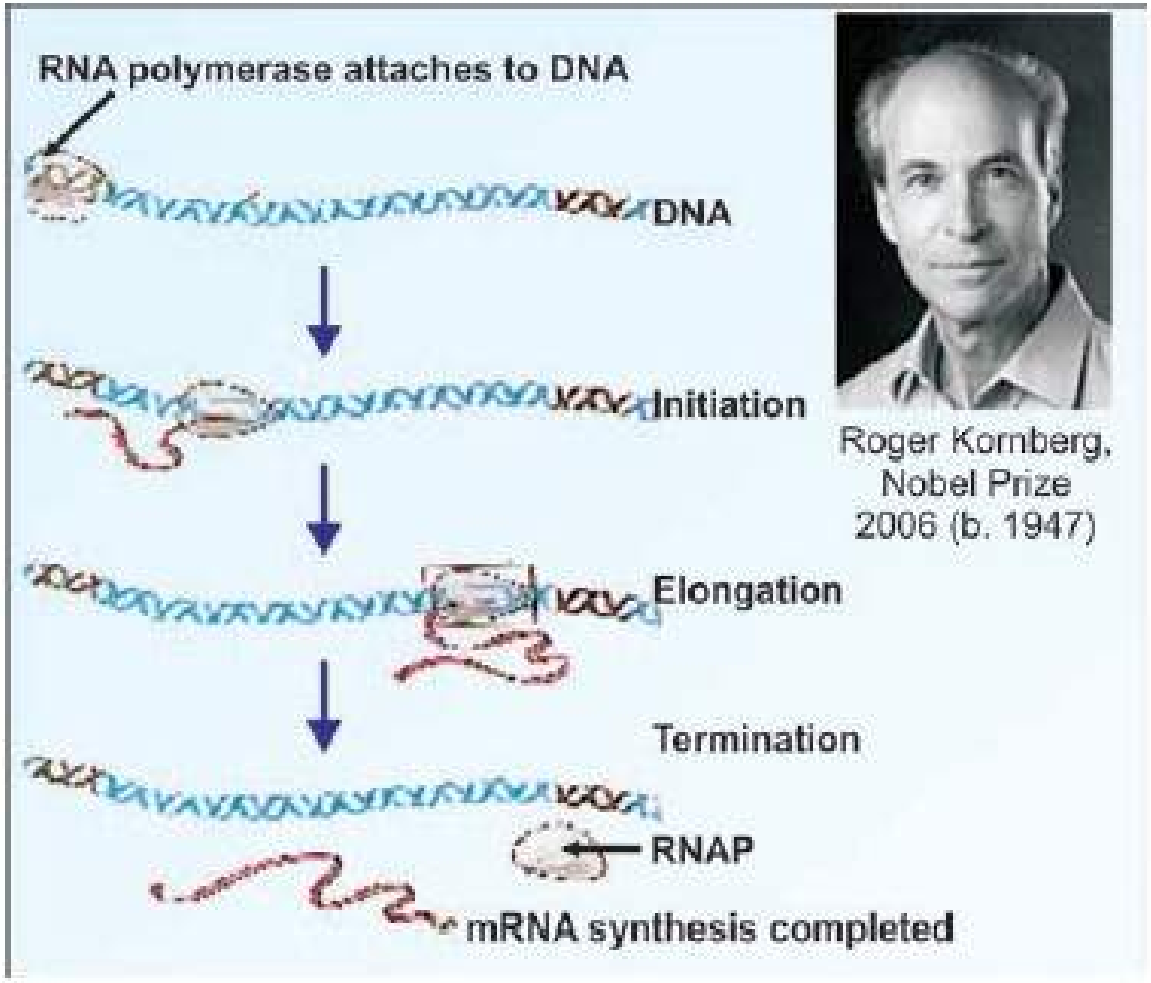



**Fig. 41.5.** Transcription process

# Reading

- Chatterjea
- Lippincott